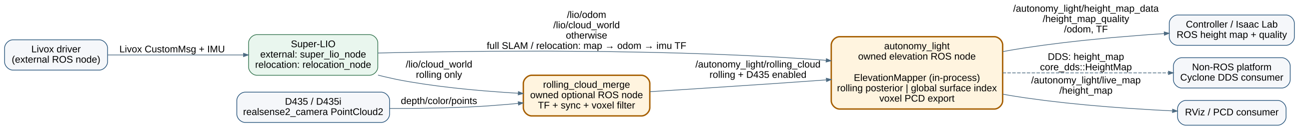
# autonomy_light software architecture

Super-LIO integration and elevation mapping · 2026-08-03

> **Design decision.** SLAM and relocalization are Super-LIO responsibilities. This package owns the elevation node and an optional, isolated D435 rolling-cloud merge node. Super-LIO remains the only SLAM backend.




## Node responsibilities

| Node | Owner | Responsibility |
|---|---|---|
| Livox driver | external | Publishes one Livox CustomMsg and IMU stream. |
| Super-LIO | external | LiDAR–IMU odometry, Scan Context + GICP + iSAM2 full SLAM, saved-map relocation, and continuous `map → odom → imu` TF. |
| `rolling_cloud_merge` | this package, optional | Synchronizes one D435/D435i PointCloud2 to a LiDAR cloud, transforms it into `odom|map`, and voxel filters it. |
| `autonomy_light` | this package | Base pose/TF, rolling or global elevation map, quality output, live PCD and PCD save. |

## Data and TF contract

| Direction | Topic | Meaning |
|---|---|---|
| input | `/lio/odom` | Super-LIO LiDAR pose; converted to `base_link`. |
| input | `/lio/cloud_world` | Registered cloud in the current global frame. |
| optional input | `/camera/.../depth/color/points` | One D435/D435i cloud; used only by the merge node. |
| intermediate | `/autonomy_light/rolling_cloud` | Time-aligned, global-frame LiDAR+D435 cloud for rolling elevation. |
| output | `/autonomy_light/height_map_data` | Controller distance grid. |
| output | `/autonomy_light/height_map_quality` | Per-cell validity, variance, age. |

| | | |
|---|---|---|
| output | `/autonomy_light/live_map` | Sparse voxel PCD for RViz and map saving. |
| optional output | `height_map` | Direct Cyclone DDS `core_dds::HeightMap` payload for a non-ROS peer. |

TF is `odom|map → base_link → base_link_gravity` and static `base_link → lidar_link`. Normal operation uses `odom`; a saved map runs Super-LIO `relocation_node` and uses `map`. After its NDT→ICP registration succeeds, Super-LIO continuously publishes `map → odom → imu`; the transform is only recalculated at the configured low global-registration rate. Mapping runs the pose graph in Super-LIO. There is no `world` frame or duplicate SLAM backend in this package.

## Elevation source selection

| Configuration | Algorithm | Use |
|---|---|---|
| `height_map.source: rolling` | Bounded Bayesian ground/upper posterior with measurement variance, aging and outlier gate. | Default for control and sim-to-real. |
| `height_map.source: global` | Persistent, PCD-derived sparse surface index. | Repeatable terrain reference from a saved or accumulated map. |

Both sources use the same output message. `valid=0` means the distance value is an unknown placeholder, not measured flat ground; Isaac Lab should consume the quality channels.

With `rolling_merge.enabled: true`, the launcher starts the optional merge node. It accepts only a D435 sample within the configured synchronization window, uses TF at that camera timestamp, and passes a LiDAR-only cloud through if TF or a matching camera sample is unavailable.

## Transport selection

`height_map_output.transport` selects `ros2`, `cyclone_dds`, or `both`. The direct DDS option uses the installed IDL `core_dds::HeightMap { sequence<float> data; }`, configurable domain and topic, and best-effort keep-last delivery. Its flat, row-major data sequence has exactly the same values and unknown convention as ROS `HeightMap.data`; geometry and quality remain ROS-message metadata. `both` is the migration/validation mode.

## Removed ownership

- Duplicate local pose graph, Scan Context loop closure, and map reconstruction
- FPFH/GICP initial localization and runtime map correction outside Super-LIO
- Secondary LiDAR merger and custom ROS-domain bridge executors
- Separate raw/refined/ROI map pipelines and duplicated path subscription

These removals make the state path linear: Super-LIO localizes; one node maps elevation.